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Experiment Title : Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

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Experiment No. 5: Demonstration of Hub Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Hub and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

Theory

1) What is a Hub?

A hub is a basic networking device used to connect multiple computers or devices in a network. It operates at the Physical Layer (Layer 1) of the OSI model. A hub does not filter or manage data; it simply connects devices and transmits data to all connected devices.

2) Working Principle of a Hub.

A hub works on the principle of broadcasting.

When a device sends data:

1. The hub receives the data signal.
2. It does not check the destination address.
3. It sends the data to all connected devices.
4. Only the intended device accepts the data; others ignore it.

Since it broadcasts data to every device, it can cause network congestion and data collisions.

3) MAC Address

A MAC (Media Access Control) address is a unique physical address assigned to a network interface card (NIC) of a device.

- It is a 12-digit hexadecimal number.
- Example: 00:1A:2B:3C:4D:5E
- It works at the Data Link Layer (Layer 2) of the OSI model.
- It is permanently assigned by the manufacturer.

It is used to identify devices within a local network.

4) IP Address

An IP (Internet Protocol) address is a logical address assigned to a device in a network to identify it and enable communication.

- It works at the Network Layer (Layer 3).
- It can be changed.
- Example of IPv4: 192.168.1.1
- Example of IPv6: 2001:0db8:85a3::8a2e:0370:7334

It helps in locating devices across different networks.

5) Local Area Network (LAN)

A Local Area Network (LAN) is a network that connects computers and devices within a small geographical area such as a home, school, office, or college.

- It allows sharing of resources like files, printers, and internet.
- It provides high-speed communication.
- It is privately managed.

6) Star Topology

A star topology is a network layout in which all devices are connected to a central device such as a hub or switch.

- Each device has a separate connection to the central device.
- If one cable fails, only that device is affected.
- If the central device fails, the entire network stops working.

It is easy to install and troubleshoot and is commonly used in LANs.

Hardware Components Required

- **PC-PT (x3):** End devices acting as network nodes.
- **Hub-PT:** A generic Physical Layer device used to interconnect the nodes.
- **Copper Straight-Through Cables:** The transmission media used for PC-to-Hub connectivity.

Hardware and Connection Setup

- **Network ID:** 192.168.1.0
- **Subnet Mask:** 255.255.255.0
- **PC Naming Convention:**
  - PC 1: **Batch_Roll_A** (e.g., S11_01_A)
  - PC 2: **Batch_Roll_B** (e.g., S11_01_B)
  - PC 3: **Batch_Roll_C** (e.g., S11_01_C)

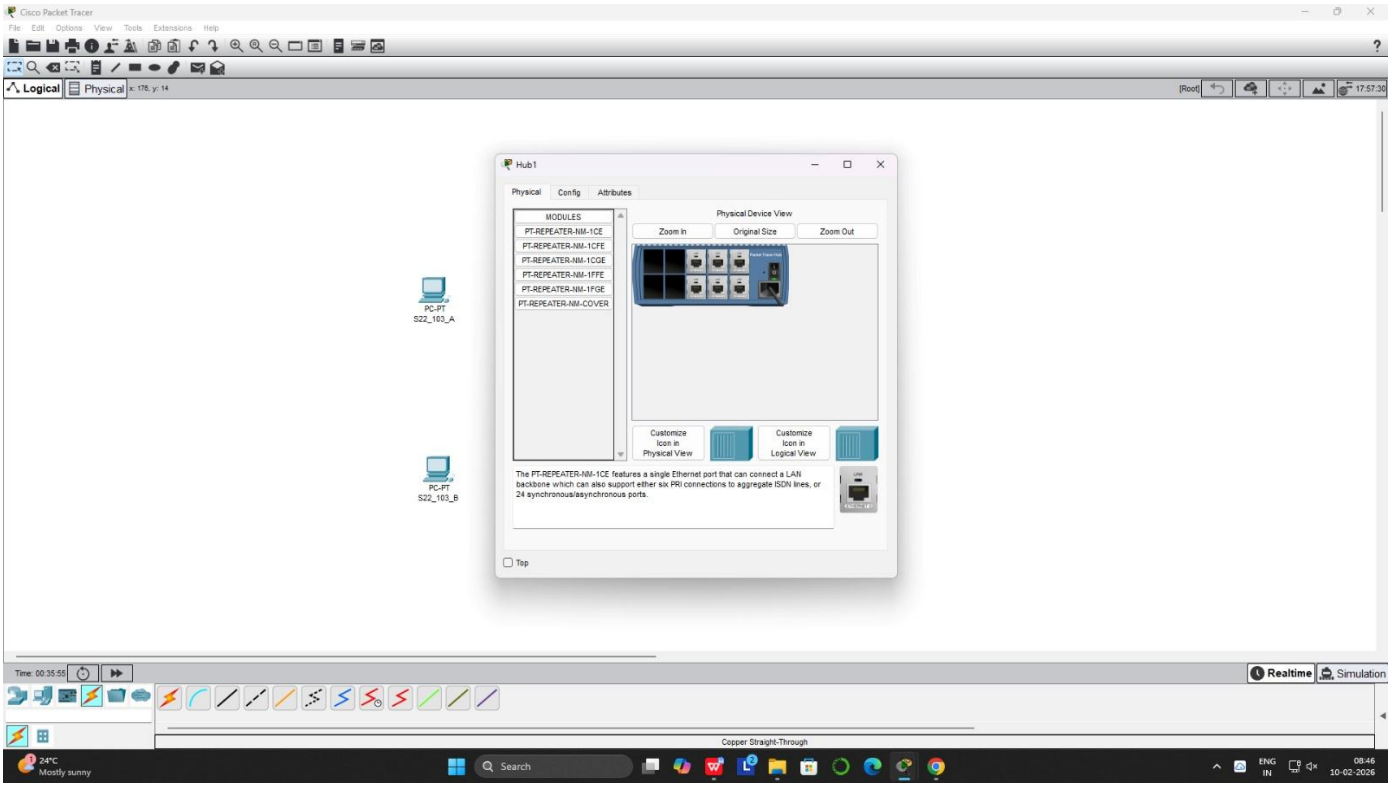
Procedure

Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select **Network Devices > Hubs > Hub-PT**. Place it in the center.
2. Select **End Devices > PC** and place four units around the Hub.
3. **Naming:** Click on each PC label and rename them following the convention: S11_01_A, S11_01_B, S11_01_C and S11_01_D.

Step 2: Hardware Identification

1. Click on the **Hub** and select the **Physical** tab.
2. Observe the ports. Note that unlike the Switch, a Hub usually has fewer ports and no "intelligence" indicators (like console ports).



Step 3: Physical Cabling

1. Select the **Copper Straight-Through** cable (solid black line).
2. Connect **S11_01_A (FastEthernet 0)** to **Hub (Port 0)**.
3. Repeat for the other three PCs using Hub ports **1, 2** and **3**.
4. **Observation:** Notice that the link lights turn **Green immediately**. Unlike a Switch, a Hub does not run STP (Spanning Tree Protocol), so there is no "Orange" waiting period.

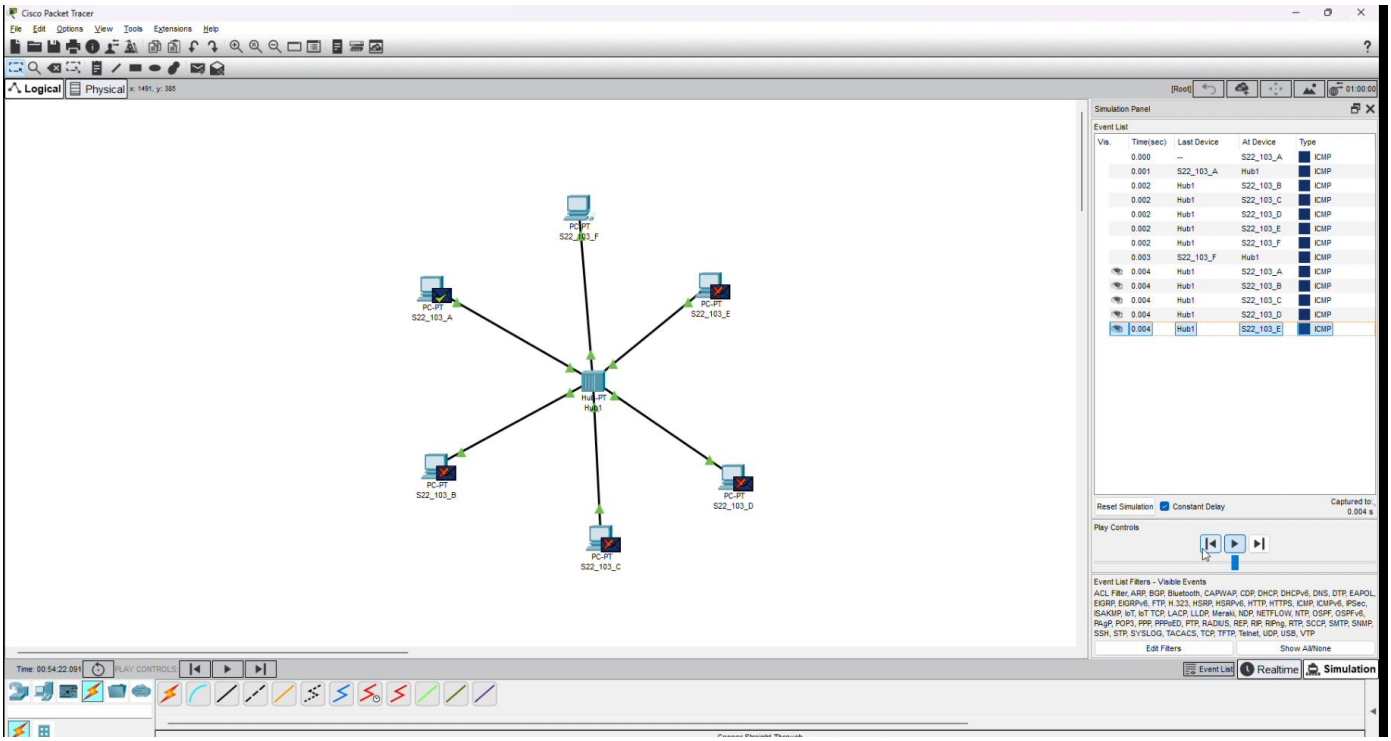
Step 4: Logical Addressing (IP Configuration)

1. Click **S11_01_A > Desktop > IP Configuration**.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure **S11_01_B** with IP 192.168.1.2, **S11_01_C** with IP 192.168.1.3 and **S11_01_D** with IP 192.168.1.4.

Step 5: Testing and Simulation (Crucial Step)

1. **Real-time:** Open the Command Prompt on **S11_01_A** and type ping 192.168.1.3.
2. **Simulation Mode:**

- Click the **Simulation** tab and filter for **ICMP**. ○ Use the **Add Simple PDU** (Envelope) to click **S11_01_A** then **S11_01_C**.
- Click **Play** and observe carefully: The packet goes to the Hub, and the Hub sends it to **all** PC B, PC C and PC D.
- Observe the **Red Cross** on PC B and PC D (because they reject a packet not meant for them) and the **Green Check** on PC C.



6. Observations

- **Instant Connectivity:** Hub ports do not have a "learning" phase; they transition to green immediately upon connection.
- **Broadcast Behavior:** During simulation, the Hub was observed to forward the packet to every connected device. This wastes bandwidth and creates security risks.
- **Device Rejection:** End devices (PCs) are responsible for checking the destination MAC address. If it doesn't match, the PC discards the packet (indicated by the red cross in simulation).

Conclusion

The experiment successfully demonstrates the hardware and functional limitations of a Hub. It concludes that a **Hub** is a non-intelligent Layer 1 device that relies on broadcasting, making it less efficient than a Switch for modern Local Area Networks.